

Quantum Track Reconstruction in the LHCb VELO

From the Hamiltonian to HHL to the 1-Bit Quantum Filter

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Quantum Tracking Mini Workshop, Maastricht — 11 June 2026

Based on TrackHHL: [arXiv:2511.11458](https://arxiv.org/abs/2511.11458) (HHL) · [arXiv:2601.07766](https://arxiv.org/abs/2601.07766) (1-Bit Quantum Filter)

Outline

- 1 Motivation & the toy model
- 2 From hits to the Hamiltonian to a matrix inversion
- 3 Quantum inversion: $\text{HHL} \rightarrow \text{one bit} \rightarrow \text{1BQF}$
- 4 Results: efficiency, false rate, and the trade-off
- 5 Attacking the survivors: the bifurcation term
- 6 Next steps: QSVT

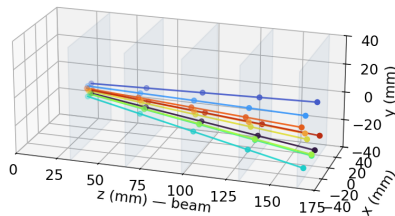
The problem: tracking is a combinatorial needle-in-a-haystack

- The LHCb **VELO** reconstructs charged particles (**tracks**) from **hits** on silicon planes.
- HL-LHC pile-up $\Rightarrow$ reconstruction cost threatens to outpace classical resources.
- With T hits per plane each plane-gap offers T^2 candidate links, only T real:

$$n_{\text{seg}} = 4T^2 \quad \text{vs} \quad n_{\text{true}} = 4T \quad (\text{signal fraction } 1/T).$$

- At $T=400$: **640,000** candidates, **1,600** true (0.25%).
- **Idea:** cast track finding as a *linear system* $A\mathbf{x} = \mathbf{b}$, solvable classically *or* on a quantum computer.

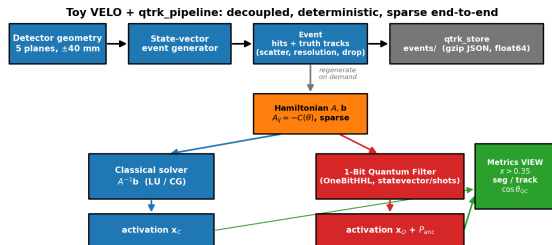
Toy VELO event: 10 tracks through 5 planes ($\phi_{\text{max}} = 0.2$)



A generated toy VELO event: tracks fan from the PV through 5 planes.

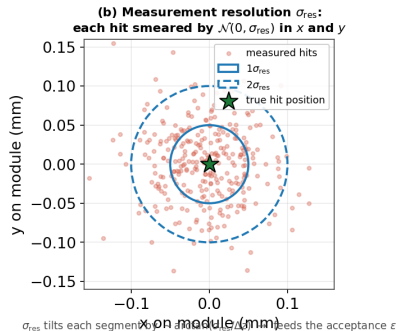
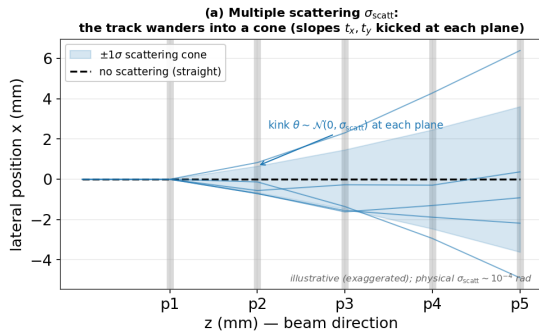
A controlled toy VELO — our testbed

- LHCb_VeLo_Toy_Model: **5 planes** at $z=33, 66, 99, 132, 165$ mm, ± 40 mm.
- Deterministic and fully controlled: sweep multiplicity T , noise, density, kernel.
- Decoupled pipeline: events stored once; the matrix A regenerated on demand (sparse); metrics recomputed as a *view*.
- Purpose: isolate *how each physical factor* moves the algorithm.



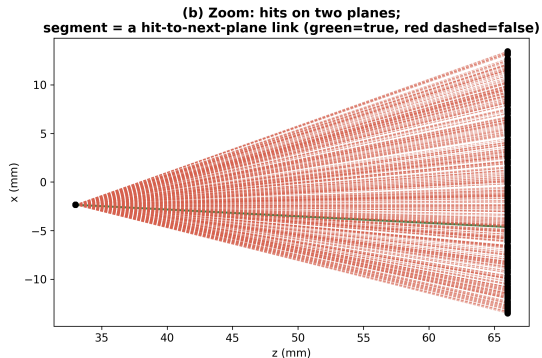
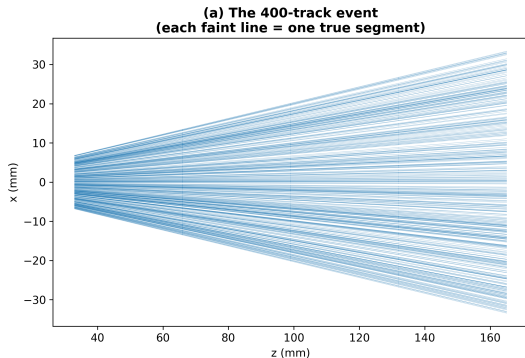
The toy injects the two physical detector noises

The two detector noises the toy injects — both widen the triplet kink angle, set by $\varepsilon = \sqrt{2(\sigma_{\text{scatt}})^2 + 12\arctan^2(\sigma_{\text{res}}/\Delta z) + \dots}$



- **Multiple scattering σ_{scatt} :** kicks the slopes (t_x, t_y) at each plane $\Rightarrow$ a cone around the ideal track.
- **Resolution σ_{res} :** smears each measured hit in x, y . Both widen the triplet kink angle $\Rightarrow$ they set ε .

Step 1: from hits to segments



- A **segment** is a candidate hit→hit link between adjacent planes; assign each a variable x_i (its *activation*).
- **Green** = true (both hits from the same particle); **red dashed** = false (cross-track accident).
- The reconstruction task: switch ON the green, OFF the red — $4T^2$ binary decisions.

Step 2: the kink-angle cost — which segments support each other

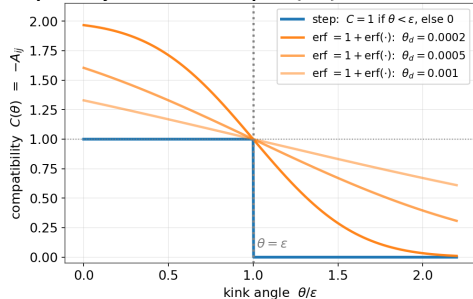
- Two segments sharing a middle hit form a **triplet** with kink angle $\theta_{ij} = \arccos(\hat{v}_i \cdot \hat{v}_j)$.
- A real particle goes nearly straight $\Rightarrow$ small θ . The **cost/compatibility kernel** is explicit:

$$C(\theta_{ij}) = \begin{cases} 1 & \text{share a hit and } \theta_{ij} < \varepsilon \\ 0 & \text{otherwise} \end{cases}$$

- smooth variant: $C = 1 + \operatorname{erf}\left(\frac{\varepsilon - \theta}{\theta_d \sqrt{2}}\right) \in [0, 2]$.
- ε is derived from the detector noise:

$$\varepsilon = \sqrt{2(\sigma_{\text{scatt}})^2 + 12 \arctan^2 \frac{s\sigma_{\text{res}}}{\Delta z} + 2\theta_{\text{min}}^2}.$$

Compatibility kernel: hard step ($\in \{0, 1\}$) vs smooth erf ($\in [0, 2]$)



Step 3: the explicit Hamiltonian

Reward mutually-compatible segments, penalise activation, bias every segment on:

$$H(\mathbf{x}) = \underbrace{\frac{\gamma + \delta}{2} \sum_i x_i^2}_{\text{activation penalty}} - \underbrace{\frac{1}{2} \sum_{i \neq j} C(\theta_{ij}) x_i x_j}_{\text{triplet reward (attractive)}} - \underbrace{\delta \sum_i x_i}_{\text{bias}}$$

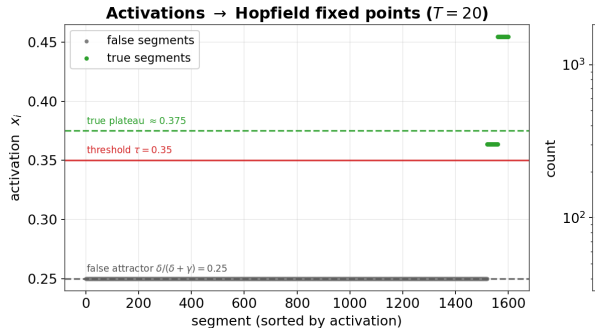
- An Ising/Hopfield-type energy over segment activations (Denby–Peterson lineage); defaults $\gamma=3$, $\delta=1$.
- Physically: each active segment “pulls up” the segments that continue it through a shared hit at a small kink angle; lone segments are pulled down.

Step 4: minimising H is a matrix inversion

Stationarity, $\partial H / \partial x_i = 0$ for every segment i :

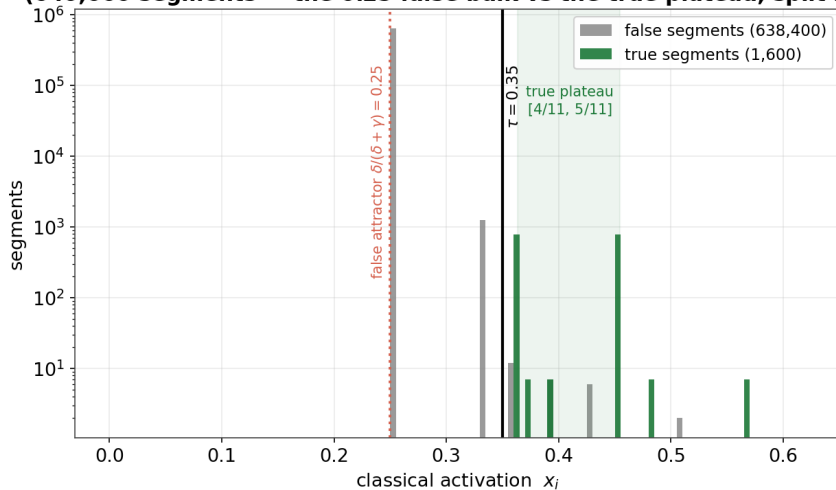
$$(\gamma + \delta) x_i - \sum_j C(\theta_{ij}) x_j - \delta = 0 \quad \Longleftrightarrow \quad \boxed{A \mathbf{x} = \delta \mathbf{1}, \quad A = (\gamma + \delta) I - C}$$

- Reconstruction = **solve a sparse linear system**; classically: $\mathbf{x} = A^{-1} \delta \mathbf{1}$ (LU/CG).
- **Hopfield fixed points** give an absolute scale: false/isolated $\rightarrow \delta / (\delta + \gamma) = 0.25$; true chain $\rightarrow [\frac{4}{11}, \frac{5}{11}]$; threshold $\tau = 0.35$.
- Classical solve = state-of-the-art efficiency [arXiv:2511.11458].



What the inversion produces on one clean 400-track event

**One clean 400-track event: the classical activation spectrum
(640,000 segments — the 0.25 false bulk vs the true plateau, split by τ)**



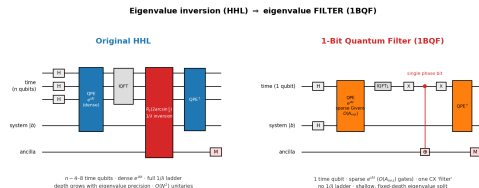
HHL: the quantum linear-system algorithm

Harrow–Hassidim–Lloyd solves $A\mathbf{x} = \mathbf{b}$:

- 1 prepare $|\mathbf{b}\rangle$ over the segment basis;
- 2 **QPE**: e^{iAt} writes each eigenvalue λ_k onto an n_{time} -qubit clock register;
- 3 controlled R_y : amplitude $\propto 1/\lambda_k$ onto an ancilla;
- 4 uncompute, post-select ancilla = 1 $\Rightarrow$ state $\propto A^{-1}\mathbf{b}$.

Exponential speedup in the number of hits — *contingent on* efficient QPE & readout.

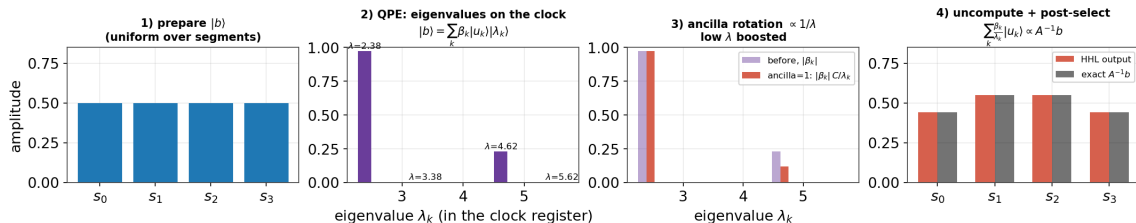
Catch: dense e^{iAt} + high-precision QPE $\Rightarrow$ prohibitive depth on NISQ hardware.



Left: the HHL circuit. Right: where we are heading.

HHL stage by stage: the statevector at every step

HHL stage by stage on a true-track block (P_4): the statevector at every step

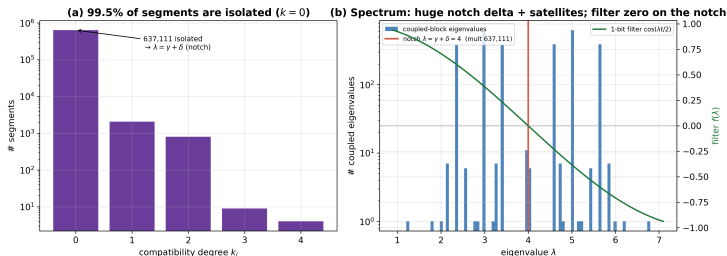


On a true-track block (P_4): $|b\rangle \xrightarrow{\text{QPE}} \sum_k \beta_k |u_k\rangle |\lambda_k\rangle \xrightarrow{R_y(1/\lambda)} \text{ancilla} = 1 \text{ branch} \xrightarrow{\text{post-select}} \propto A^{-1}b$.

TrackHHL: restrict the phase estimation to one bit

[arXiv:2511.11458 — Chiotopoulos, Lucio Martinez, Nicotra *et al.*]

- For this Hamiltonian the spectrum is **bimodal**: a huge bulk at $\lambda = \gamma + \delta$ (lone/false segments) and a narrow band of track modes away from it.
- One bit of eigenvalue information — *on the bulk or off it* — is already the answer.
- Restricting QPE to **one clock qubit**: cuts circuit depth by up to 10^4 ; resolves the read-out problem (the output is binary); enables primary-vertex post-processing.



The event spectrum: 637k eigenvalues exactly at $\gamma + \delta$ (the false bulk) + coupled satellites.

The 1-Bit Quantum Filter: the maths

[arXiv:2601.07766 — Chiotopoulos, Nicotra, Scriven *et al.*]

Choose the evolution time $t = \pi/(\gamma + \delta)$. One-qubit QPE on e^{iAt} leaves, per eigencomponent,

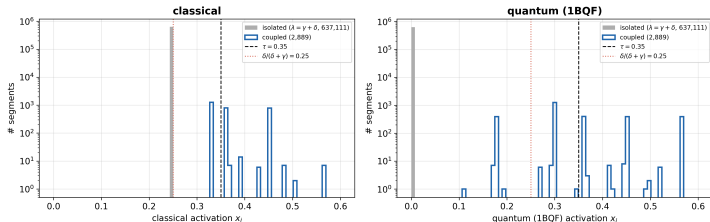
$$|u_k\rangle \left(\cos \frac{\lambda_k t}{2} |0\rangle_{\text{clock}} - i \sin \frac{\lambda_k t}{2} |1\rangle_{\text{clock}} \right),$$

and the X--CX--X ancilla flip + post-selection keeps each mode with weight

$$w_Q(\lambda) = \cos \left(\frac{\pi}{2} \frac{\lambda}{\gamma + \delta} \right)$$

$w_Q(\gamma + \delta) = 0$ (the notch, placed *exactly* on the false bulk).

Notch rotation: classical leaves the isolated bulk at 0.25; the 1BQF sends it to 0

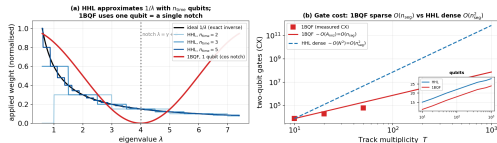


The 0.25 isolated-false bulk (637k segments) is rotated to 0; the true band survives.

Why the 1BQF wins — in brief

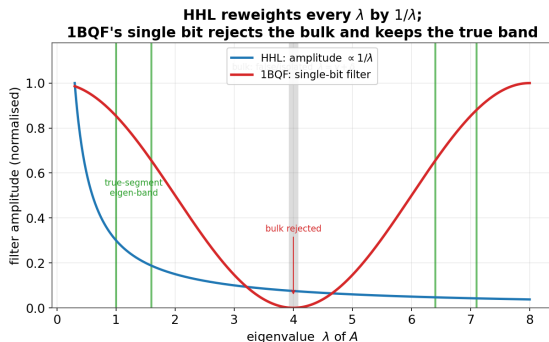
- **One clock qubit** instead of an n_{time} -qubit eigenvalue register.
- **Depth** cut by up to 10^4 vs full HHL.
- **Sparse** e^{iAt} (two-level/Givens decomposition) $\Rightarrow$ gate complexity $O(\sqrt{N} \log N)$.
- **Read-out solved**: the answer is binary (on/off the notch), no amplitude tomography.
- **NISQ-validated**: Quantinuum H2 and IBM Heron noise models.
- **Enables PV reconstruction** as classical post-processing.

Why one bit: HHL needs an n_{time} -qubit eigenvalue register and a dense e^{iAt} ; the 1BQF needs one qubit + a sparse e^{iAt} , and the single notch already separates the bimodal track spectrum



The conceptual reformulation: inversion $\rightarrow$ filtering

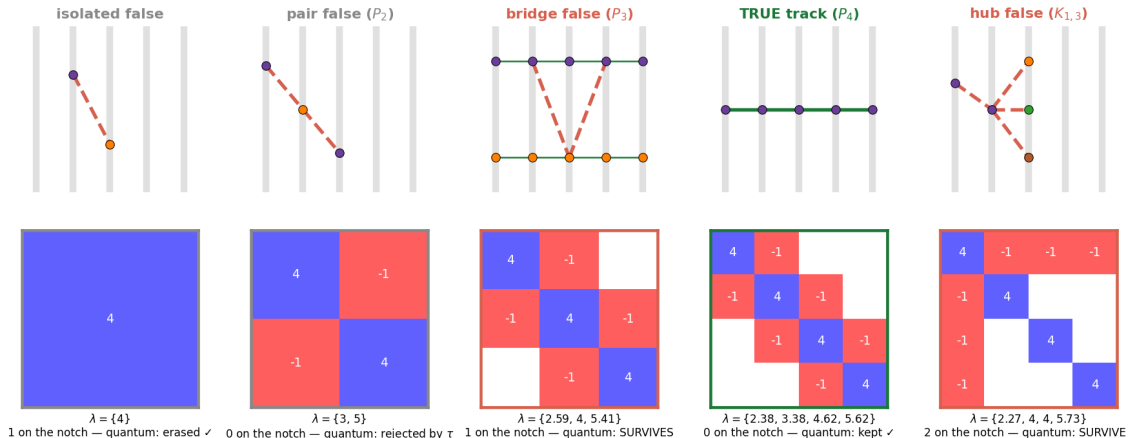
- We never needed the *numbers* in $A^{-1}\mathbf{b}$ — only a **binary decision** per segment: on or off.
- The spectrum is **bimodal**: false bulk at $\gamma + \delta$, true band away from it.
- So **one bit of spectral information classifies**: “matrix inversion” becomes “**project out the known-bad eigenspace**”.
- HHL’s eigenvalue *precision* was wasted effort for this problem — the 1BQF spends one bit exactly where the physics puts the information.



$1/\lambda$ reweighting (HHL) vs the single-notch filter (1BQF).

Every cluster type in segment space — and its block of A

Each cluster in segment space (top) and its block of A (bottom): $A = (\gamma + \delta)I - C$ is block-diagonal over clusters

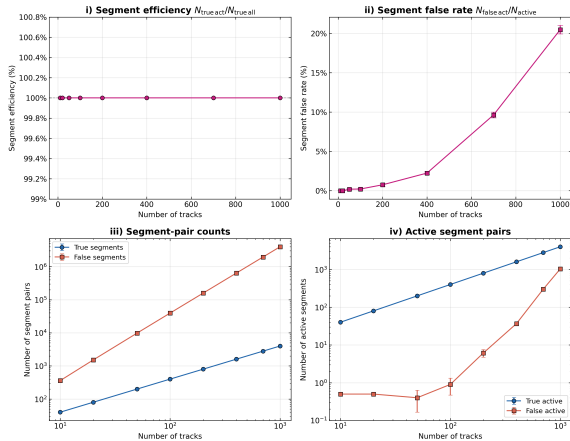


green solid = true segment · red dashed = false segment · dot colour = which real track the hit belongs to

Results: segment efficiency stays high; the false rate grows

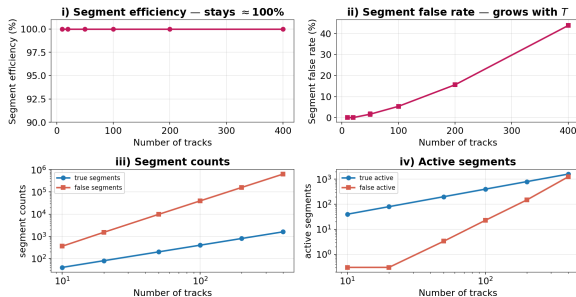
Classical

Classical solver — segment metrics ($\epsilon=2$ mrad, $\gamma=3$, no hit drop)



1BQF at matched efficiency

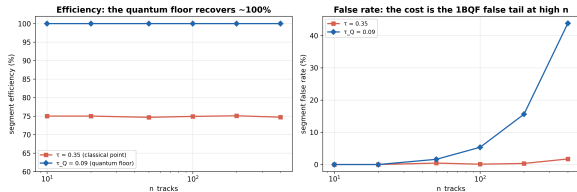
Quantum (1BQF) segment metrics at the MATCHED-EFFICIENCY operating point (full-norm rescale at $\tau=0.35 \leftrightarrow$ lowering T_0 : efficiency = 100%, false rate grows)



Both solvers: efficiency $\approx 100\%$ across multiplicity; the false rate grows with T (classical $0 \rightarrow 21\%$ by $T=1000$; quantum $0 \rightarrow 44\%$ by $T=400$) — the coupled cross-track clusters.

The efficiency–false-rate trade-off

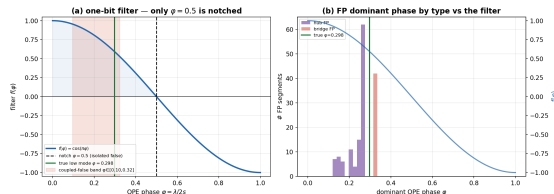
Operating-point trade-off — both achievable up to moderate T , then the false tail bites



- Recovering *every* true segment forces the threshold into the false band: **high efficiency and high false rate come together**.
- Quantum at matched efficiency ($\approx 100\%$): false rate $\rightarrow 44\%$ at $T=400$.
- The alternative fixed- τ point (eff 75%, far 1.7%) only *hides* the trade-off by discarding a quarter of the true segments.
- The **operating curve**, not a single point, is the result.

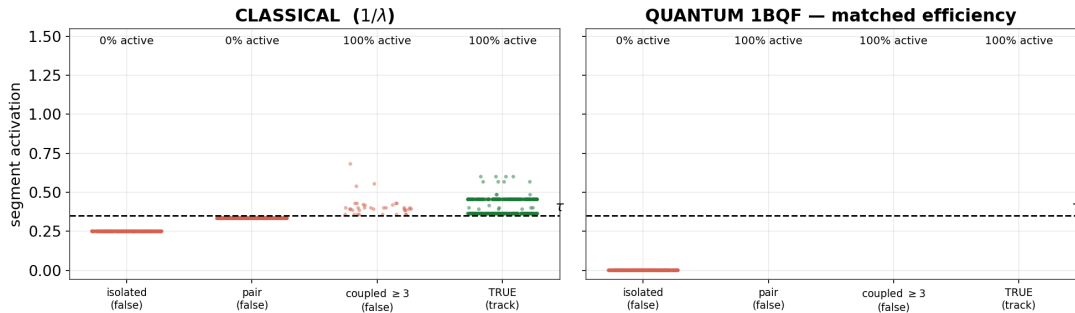
Why the trade-off is fundamental

- The notch removes only the **isolated** false class (it *is* the bulk eigenvalue).
- Surviving false positives = **coupled cross-track bridges/hubs**: *good*, off-notch eigenvalues.
- In QPE phase they fall in the **same band as true segments** — one zero cannot separate them.
- **Topological degeneracy**: a false chain shaped like a true track has an *identical* A -block $\Rightarrow$ no threshold, classical or quantum, can split them.



Class by class at matched efficiency ($T=400$)

Matched-efficiency operating point ($T = 400$): the quantum keeps ALL true segments — and with them the coupled false clusters (high efficiency $\Rightarrow$ high false rate)



quantum keeps **all** true segments — and with them the coupled false clusters: the price of efficiency.

The

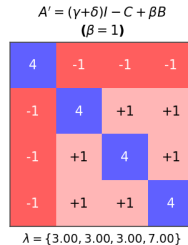
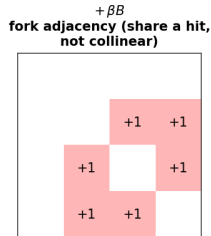
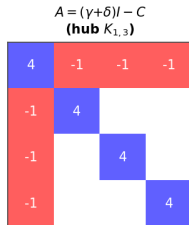
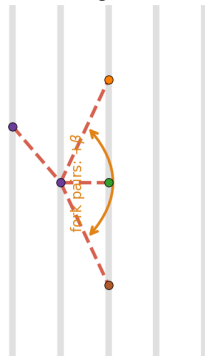
Add a bifurcation (fork) penalty to the Hamiltonian

The survivors are **competing continuations** — segments fanning from a shared hit. Penalise them:

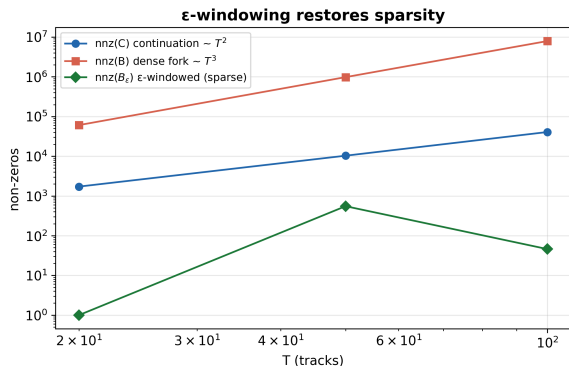
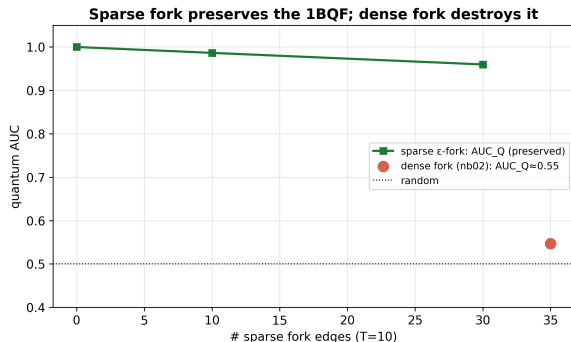
$$A' = (\gamma + \delta)I - C + \beta B, \quad \mathbf{b} = \delta \mathbf{1},$$

with B the **fork adjacency**: segments sharing a start/end hit that are *not* collinear continuations. Every segment touches exactly two hits $\Rightarrow$ the diagonal stays constant $\Rightarrow$ **still 1BQF-compatible**.

The bifurcation term penalise competing continuations — ϵ -windowed B_ϵ keeps it sparse and 1BQF-safe
 a fork: competing continuations sharing one hit



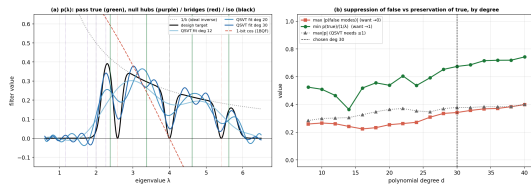
The fork term: dense breaks the quantum — ε -windowed fixes it



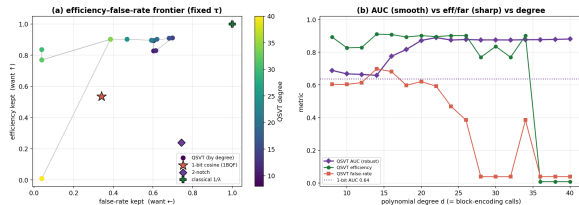
- **Dense fork** ($O(T^3)$ pairs): classically a benign uniform down-scaling ($\text{AUC} \approx 1$) but **destroys the 1BQF** ($\text{AUC} \rightarrow 0.55$) and the sparsity ($\sim 20\times$ slower).
- **ε -windowed B_ε** (only near-collinear competitors): **sparse, targeted** (false rate $\rightarrow 0$ at $\sim 2\%$ efficiency cost), and **1BQF-safe** ($\text{AUC } 0.96\text{--}0.99$).

Beyond one notch: Quantum Singular Value Transformation

- The 1BQF is QSVT restricted to $p = \cos$ — **one** notch.
- **QSVT** (Gilyén–Su–Low–Wiebe) realises *any* bounded polynomial $p(A)$ from a block-encoding of $A + d$ phase rotations.
- Target: a **band-limited inverse** — $p(\lambda) \approx 1/\lambda$ on true-track eigenvalues, $p(\lambda) \approx 0$ on every false mode (*many* notches).
- Works *iff* false and true eigenvalues are separated — on real $T=200$ events the false modes sit ≥ 0.2 from every true eigenvalue.



QSVT: a polynomial filter lifts true/false separation



- A degree- ~ 30 polynomial lifts true/false **AUC** from 0.49 (classical) / 0.64 (1-bit) to **0.88**.
- Nulls *all* off-length bridges and $\sim 95\%$ of hubs.
- **Cost:** ~ 30 block-encoding calls $\times \sim 2\times$ amplitude-amplification $\approx \text{tens} \times$ the 1-bit depth; sparsity preserved.
- **Irreducible floor:** a false bridge of the *same length* as a true track (0% at $T \leq 200$; grows with density).

Summary

- Hits $\rightarrow$ segments $\rightarrow$ explicit kink-angle cost $C(\theta) \rightarrow$ Ising-like $H(\mathbf{x}) \rightarrow$ stationarity $\Rightarrow$ **matrix inversion** $A\mathbf{x} = \delta\mathbf{1}$, $A = (\gamma + \delta)I - C$.
- **HHL** inverts it but needs deep circuits; **TrackHHL** restricts QPE to one bit (10^4 depth cut); the **1BQF** reframes inversion as a **single-notch spectral filter** — $O(\sqrt{N} \log N)$, NISQ-ready.
- On the toy: **efficiency stays $\approx 100\%$, the false rate grows with density** — a fundamental trade-off set by topologically-degenerate cross-track clusters.
- Hamiltonian-level fix: the ε -**windowed fork term** (targeted, sparse, 1BQF-safe). Algorithm-level next step: **QSVT** polynomial filters (AUC $\rightarrow 0.88$).

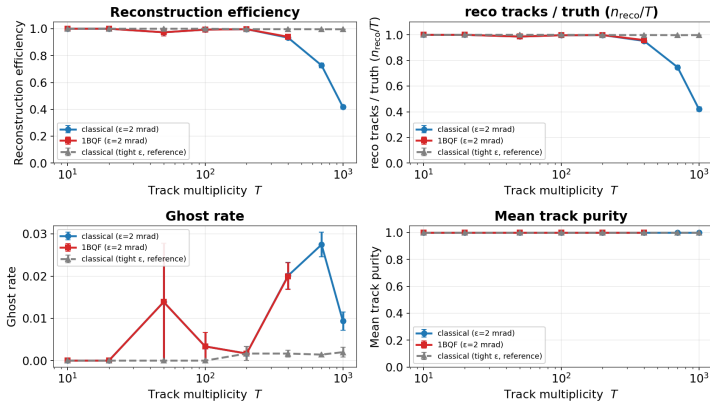
arXiv:2511.11458 (HHL) · arXiv:2601.07766 (1BQF) · toy: LHCb_VeLo_Toy_Model

Thank you

Questions?

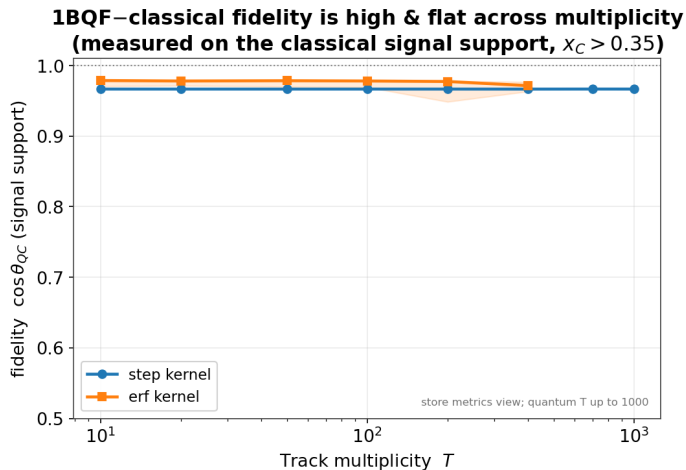
Backup: track-level metrics — segments propagate to tracks

Track-level LHCb metrics: segment false-positive bridges merge tracks at high T
(wide $\epsilon=2$ mrad collapses efficiency; tight ϵ stays perfect — same events)



Wide $\epsilon \Rightarrow$ bridges merge tracks $\Rightarrow$ reconstruction efficiency collapses ($1.0 \rightarrow 0.42$); tight ϵ stays perfect.

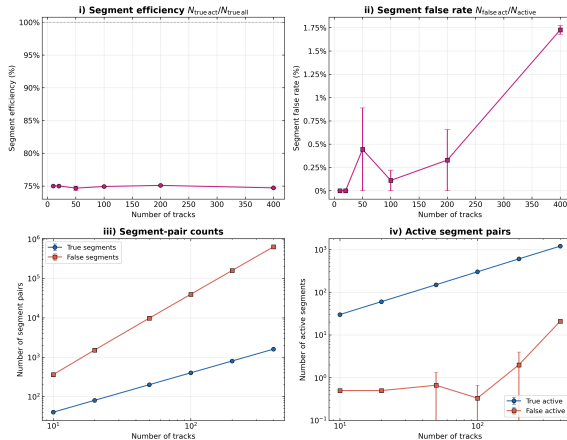
Backup: 1BQF—classical fidelity vs multiplicity



On the classical signal support the 1BQF is faithful: $\cos \theta_{QC} \approx 0.97$, flat to $T=1000$.

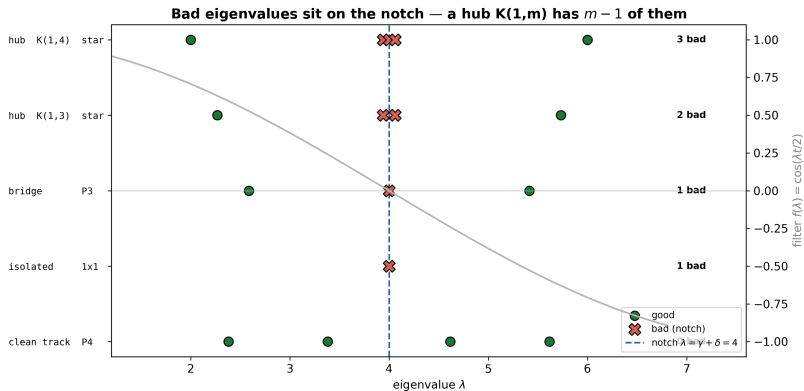
Backup: fixed-threshold operating point ($\tau = 0.35$)

Quantum (1BQF) solver — segment metrics ($\epsilon=2$ mrad, $\gamma=3$, no hit drop)



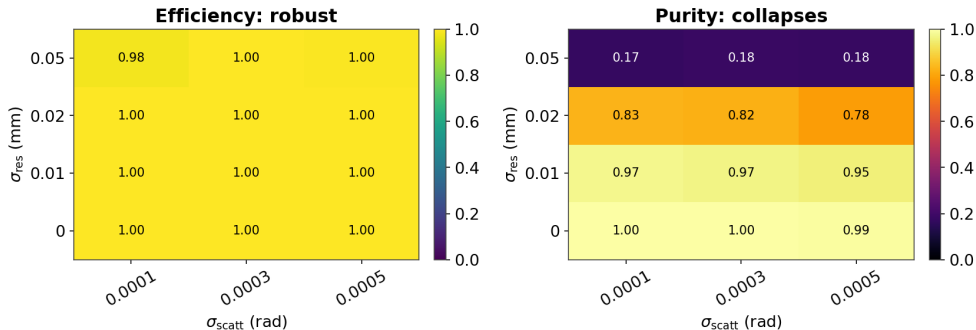
The same data at fixed τ : eff 75%, far 1.7% — the conservative end of the same operating curve.

Backup: the bad-eigenvalue ladder



Backup: classical failure mode under noise

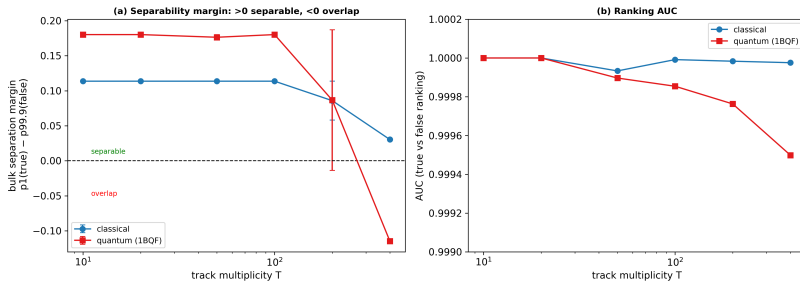
Classical failure mode at $T = 200$: resolution smearing spares efficiency but collapses purity



ε scales with noise $\Rightarrow$ efficiency robust, but **purity** collapses under resolution smearing.

Backup: separability vs multiplicity

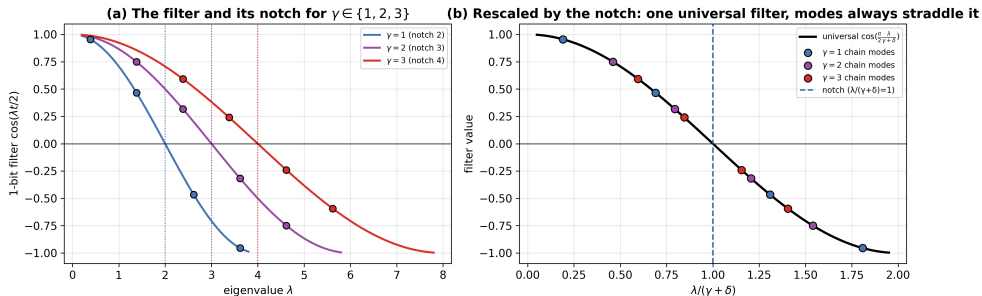
Separability vs track multiplicity (fixed- ϵ , $\gamma=3$, drop=0): quantum starts cleaner, degrades faster



Quantum starts cleaner, degrades faster; margin negative $\sim T=400$ — the coupled-false ceiling.

Backup: γ self-similarity of the notch

Self-similarity in γ : the chain spectrum is centred on $\gamma + \delta$, so the 1-bit notch always splits true segments into two plateaus



Backup: classical vs quantum per cluster (analytic)

